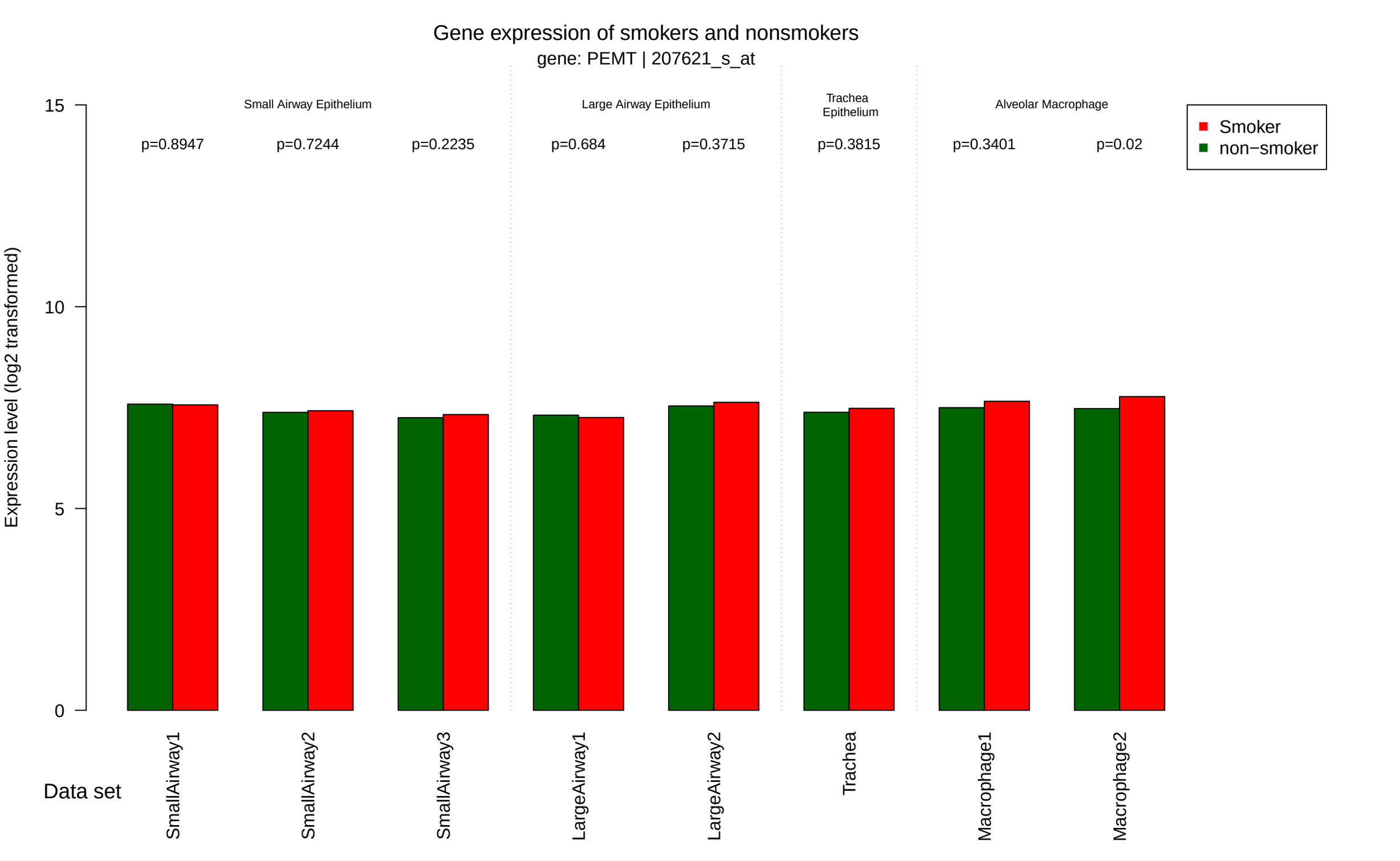
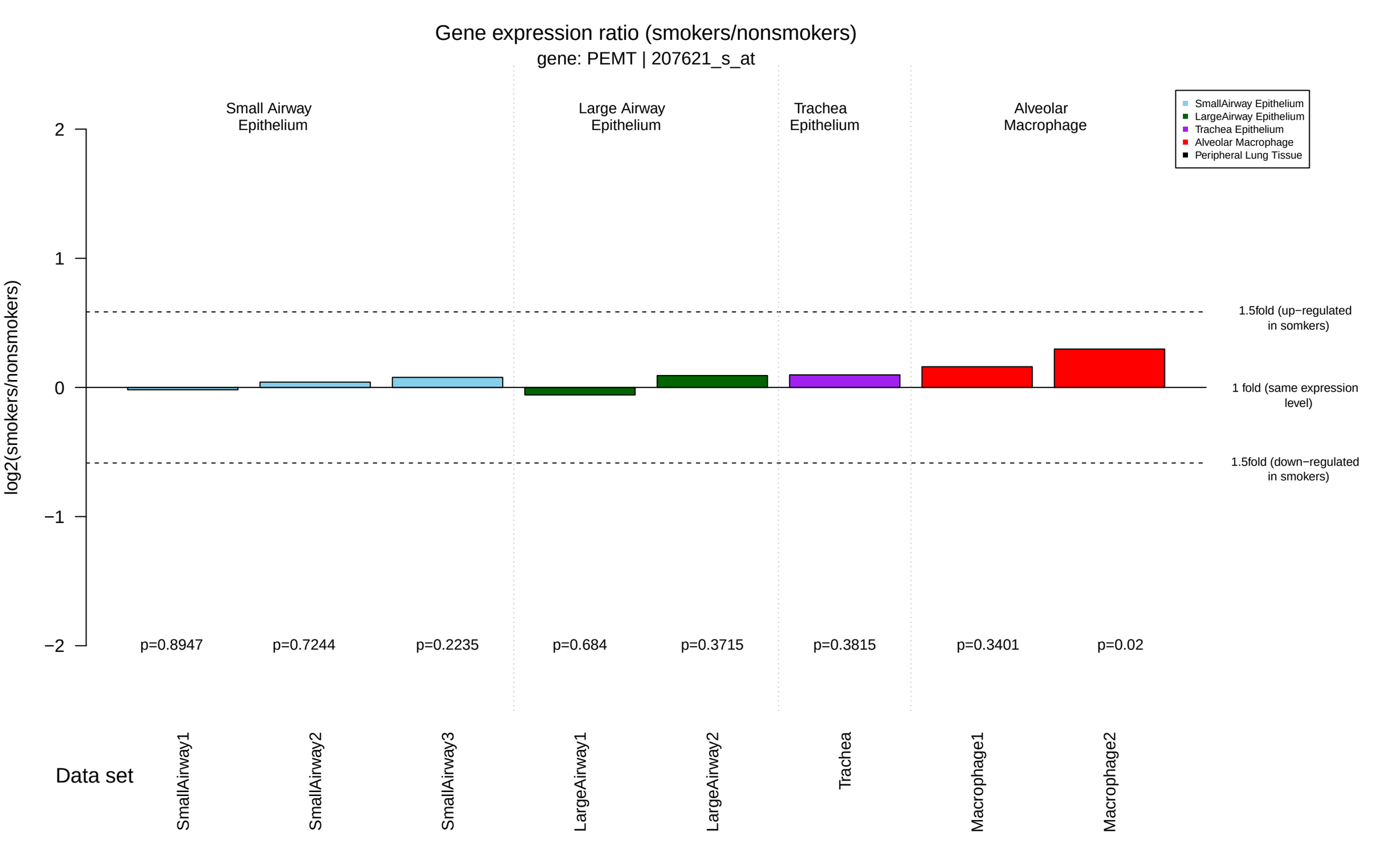
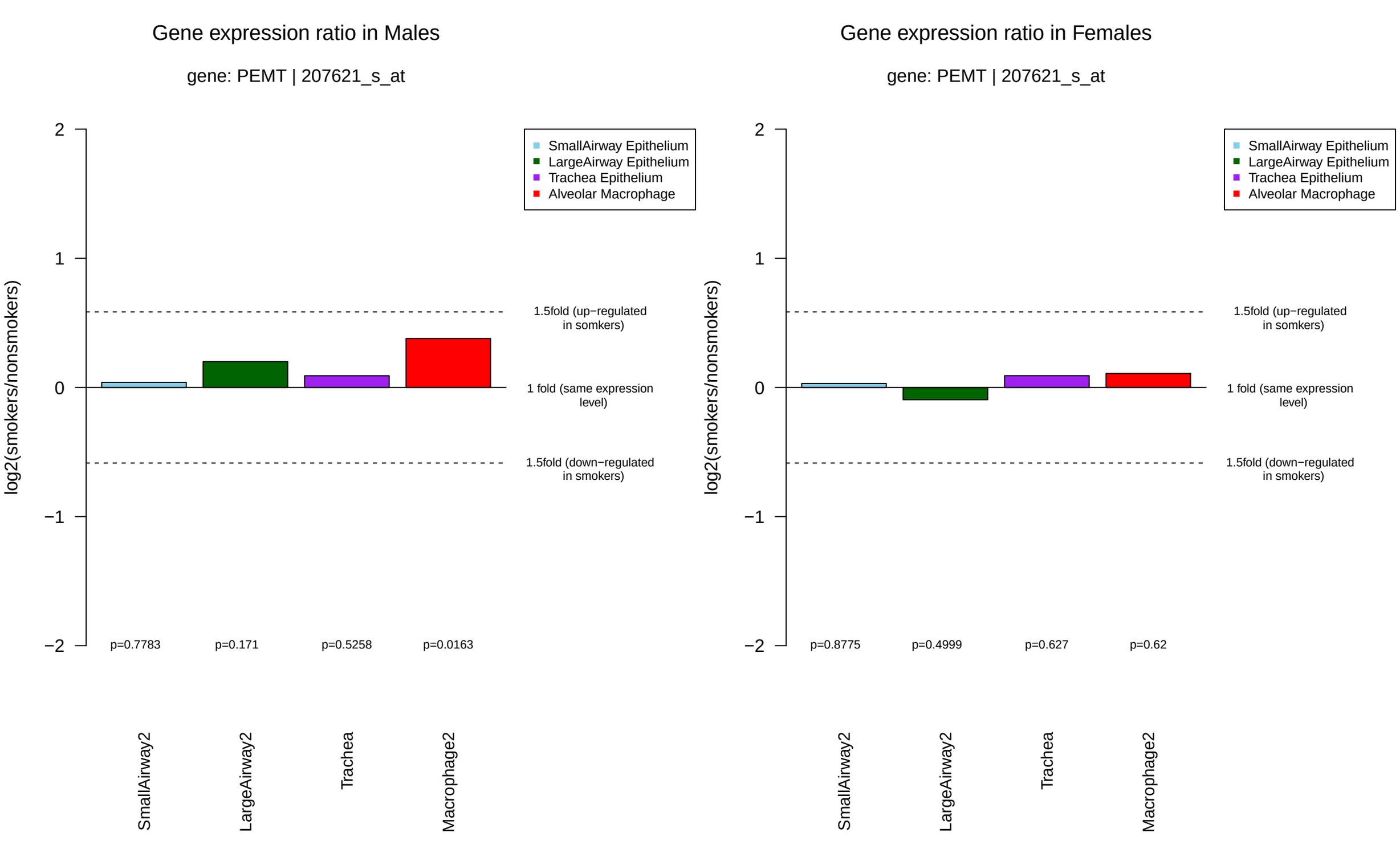
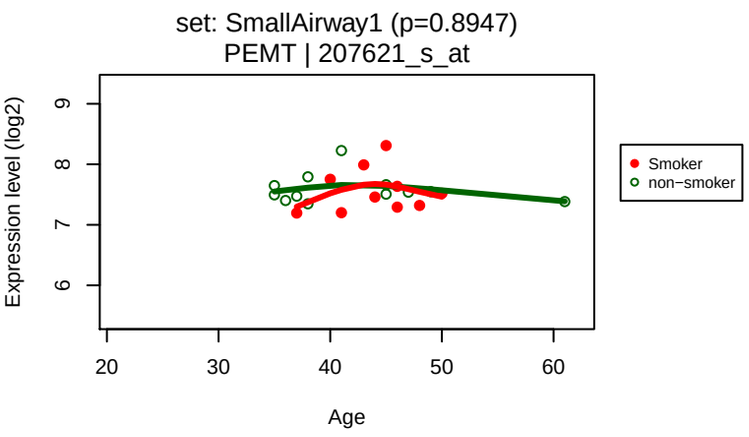
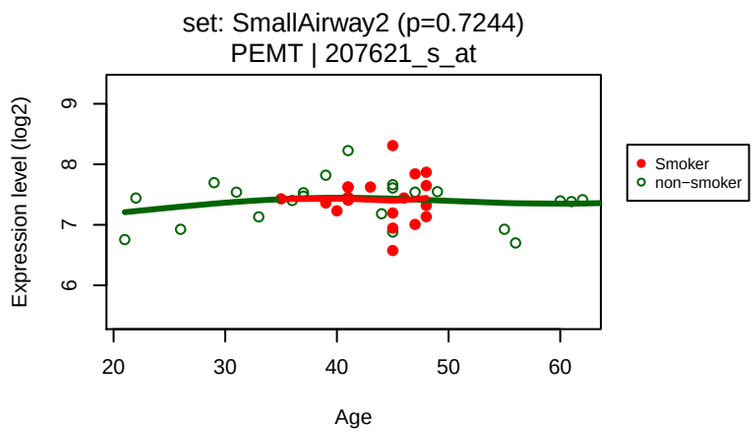
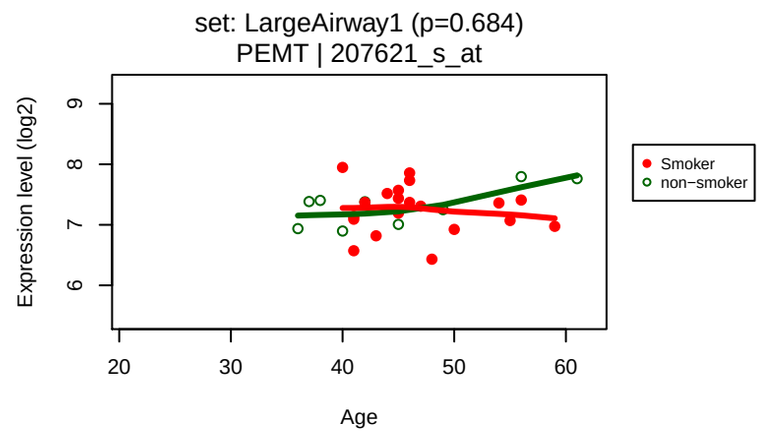

Gene expression of smokers and nonsmokers
gene: PEMT | 207621_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8947
p=0.7244
p=0.2235
p=0.684
p=0.3715
p=0.3815
p=0.3401
p=0.02
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

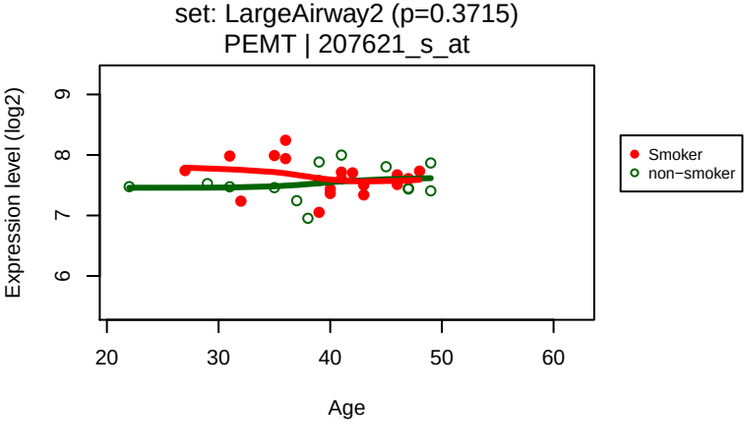

Gene expression ratio (smokers/nonsmokers)
gene: PEMT | 207621_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8947
p=0.7244
p=0.2235
p=0.684
p=0.3715
p=0.3815
p=0.3401
p=0.02
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

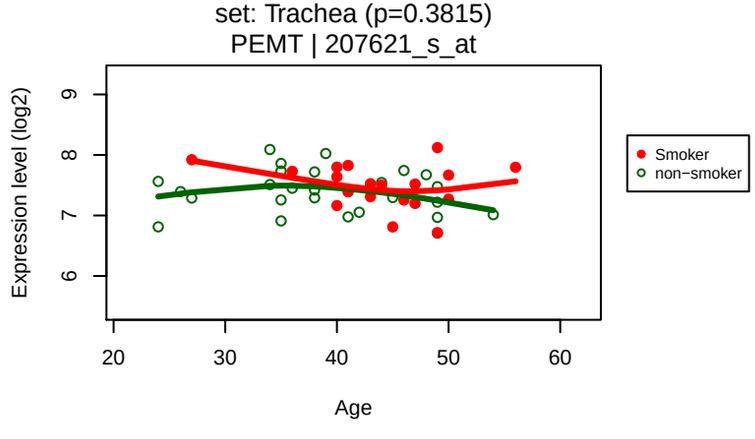

Gene expression ratio in Males
gene: PEMT | 207621_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7783
p=0.171
p=0.5258
p=0.0163
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PEMT | 207621_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8775
p=0.4999
p=0.627
p=0.62
SmallAirway2
LargeAirway2
Trachea
Macrophage2

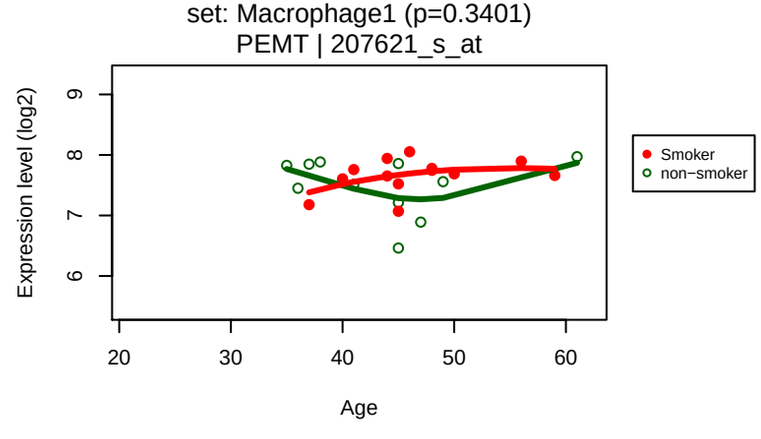

set: SmallAirway1 (p=0.8947)
PEMT | 207621_s_at
Expression level (log2)
Age
Smoker
non−smoker

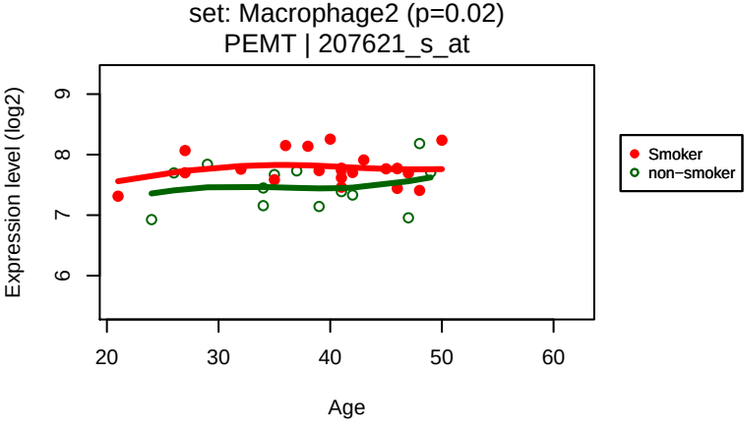

set: SmallAirway2 (p=0.7244)
PEMT | 207621_s_at
Expression level (log2)
Age
Smoker
non−smoker

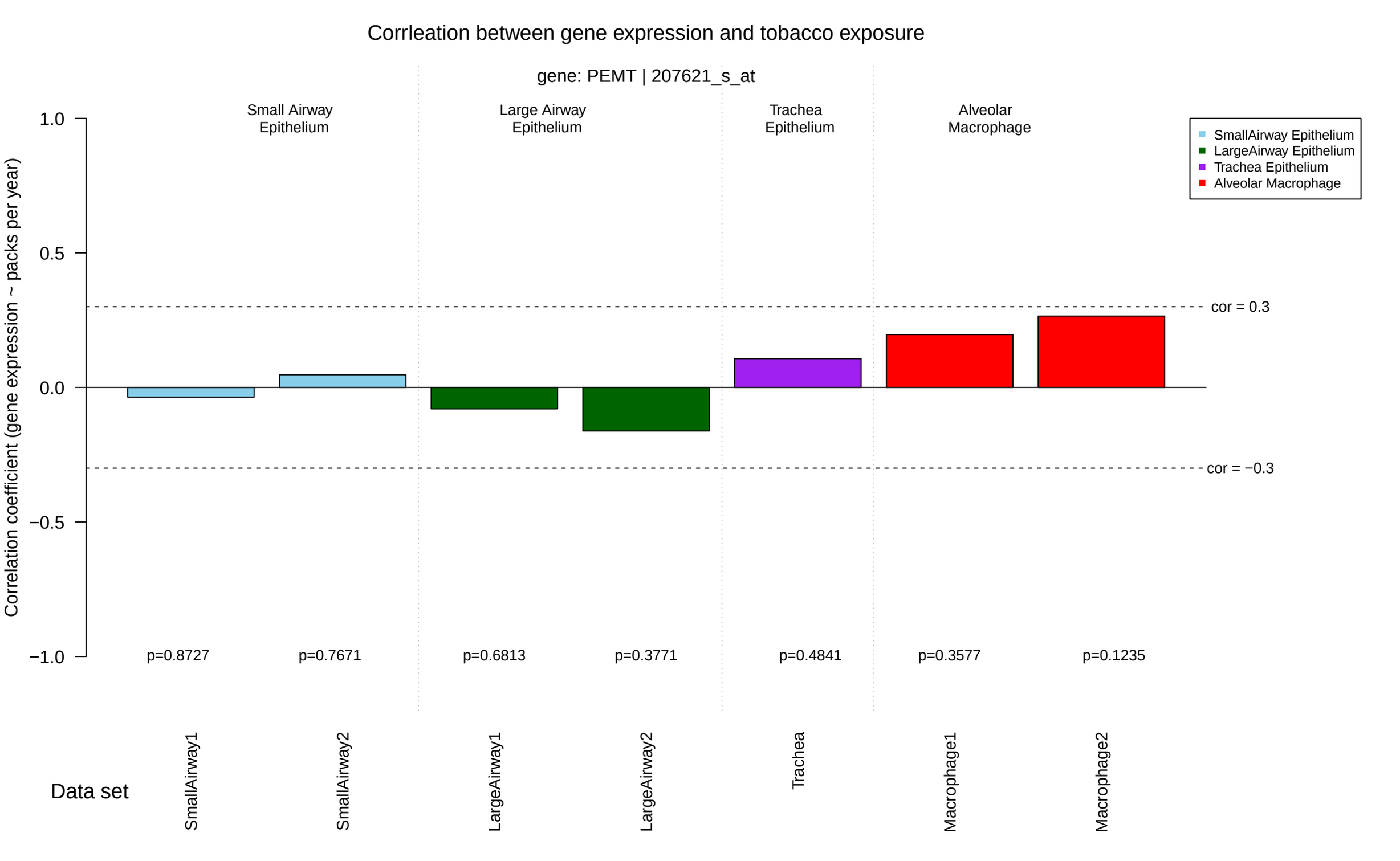

set: LargeAirway1 (p=0.684)
PEMT | 207621_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3715)
PEMT | 207621_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3815)
PEMT | 207621_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3401)
PEMT | 207621_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.02)
PEMT | 207621_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PEMT | 207621_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8727
p=0.7671
p=0.6813
p=0.3771
p=0.4841
p=0.3577
p=0.1235
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2